

Phase 7 — Automation Suite and Runbooks

Harrington Capital plc | Infrastructure Engineer Simulation

Objective

Build the automation suite that prevents repeat incidents and saves the team hours of manual work every week. Scripts run. Output captured. Runbooks filed. Everything committed.

By the end of this phase you will have: - PowerShell disk health monitor running and producing output - PowerShell bulk user provisioning script tested with real CSV - Ansible patch baseline applied to all Linux hosts - Two runbooks committed to `/runbooks`

BUILD

Task A — PowerShell Disk Health Monitor

Build

On HC-DC01, run:

```
cd C:\path\to\harrington-infrastructure-operations\powershell
.\03-disk-health-check.ps1
```

The script checks all servers in its `$Servers` array and reports: - Drive, total GB, used GB, free GB, used %, status (OK / WARNING / CRITICAL)

Verify

Output must show: - At least 3 servers checked - Status column (OK, WARNING, or CRITICAL) - Log file written to `C:\HC-Logs\`

Submit

Paste the `DISK HEALTH SUMMARY` table from the console output.

Task B — PowerShell Bulk User Provisioning

Build

First, create a CSV file of new starters. Copy the template from `/evidence/new-starter-template.csv` and add 3 test users.

Example content:

```
FirstName,LastName,Department,Title,Manager
Test,User1,Technology,Junior Engineer,j.okafor
Test,User2,Finance,Finance Analyst,f.oduya
Test,User3,Operations,Operations Assistant,c.mwangi
```

Save as `C:\HC-Logs\new-starters.csv` on HC-DC01.

Run the provisioning script:

```
.\04-user-provisioning.ps1 -CsvPath "C:\HC-Logs\new-starters.csv"
```

Verify

```
# Confirm users were created
Get-ADUser -Filter "Name -like 'Test*'" | Select-Object Name, SamAccountName, Enabled

# Clean up test users after verification
Get-ADUser -Filter "Name -like 'Test*'" | Remove-ADUser -Confirm:$false
```

Submit

Paste the `PROVISIONING SUMMARY` from the console output. Include the count: `Created: 3 | Failed: 0`.

Task C — Ansible Patch Baseline

Build

On your Ansible control machine:

```
cd harrington-infrastructure-operations/ansible
ansible-playbook -i inventory.ini patch-baseline.yml
```

This patches all Linux hosts one at a time (serial: 1) to reduce blast radius.

Verify

```
# Check patch log on HC-APP01
ssh hcadmin@<HC-APP01-IP> "cat /var/log/hc-patch-log.txt"

# Confirm no pending security updates
ssh hcadmin@<HC-APP01-IP> "apt list --upgradable 2>/dev/null | grep -c security"
# Should be 0
```

Submit

- Full Ansible PLAY RECAP
- Contents of `/var/log/hc-patch-log.txt` from each host

Task D — Grafana Disk Alert Rule

Add an alert to Grafana that fires when disk usage exceeds 75%:

1. Open Grafana → Alerting → Alert rules → New alert rule
2. Name: `HC Disk Usage Warning`
3. Query:

```
100 - ((node_filesystem_avail_bytes{mountpoint="/" } / node_filesystem_size_bytes{mountpoint="/" }) * 100)
```

1. Condition: IS ABOVE 75
2. Evaluation: every 1m for 5m
3. Save

Screenshot the alert rule configuration.

Task E — Verify Runbooks

The runbooks already exist in `/runbooks/`. Your task is to:

1. Read both runbooks end-to-end.
2. Follow RB-USER-PROVISIONING-v1 for your Task B above — document that you followed it.
3. Add a brief entry at the bottom of each runbook confirming it was tested:

```
---

## Test Record

Date: 2026-06-09

Tested by: [Your name]

Result: PASS

Notes: All steps completed successfully. Script produced correct output.
```

Commit the updated runbooks:

```
git add runbooks/
git commit -m "docs(runbooks): test records added after Phase 7 verification"
git push origin main
```

VERIFY

```
# PowerShell scripts ran successfully
ls C:\HC-Logs\disk-check-*.log # Should exist
ls C:\HC-Logs\user-provisioning-*.log # Should exist

# Ansible patched successfully
ssh hcadmin@<HC-APP01-IP> "cat /var/log/hc-patch-log.txt"

# Grafana alert exists
curl -s http://admin:admin@localhost:3000/api/v1/provisioning/alert-rules | python3 -m json.tool | g
```

SUBMIT

#	Evidence	How to get it
1	Disk health check output	Console DISK HEALTH SUMMARY table
2	User provisioning output	Console PROVISIONING SUMMARY (Created: 3, Failed: 0)
3	Ansible PLAY RECAP	Full recap showing both hosts patched
4	Grafana alert rule screenshot	Alert rule configuration page
5	Runbook GitHub URLs	Both runbooks with test records committed

Anti-fake check: Disk health log file must exist at `C:\HC-Logs\` with today's date in the filename. Ansible log must show real timestamps on each host.